

ZS-T10

The $T_Y \rightarrow X$ Sector-Conversion Operator:

A Geometry-First ANC Correction $\delta \ln \gamma_\square = -\kappa^2$ for ${}^3\text{He}(\alpha,\gamma){}^7\text{Be}$, a Per-Nucleus Falsification Programme, and the ${}^8\text{B}$ Consistency Test

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Verification: 32 checks PASS | Gate Li-2 FLAGGED (nonlinear ${}^7\text{Li}$ tension) | Zero Free Parameters

$A = \delta X \cdot \delta Y = (5/19)(7/23) = 35/437$ (polyhedral, ZS-F2, BBN-independent); $Q = 11$; $\kappa^2 = A/Q$

Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem; standard math alone, machine-verifiable.
DERIVED-CONDITIONAL	DERIVED conditional on a listed axiom set or upstream theorem.
HYPOTHESIS-strong	Structurally motivated conjecture with corpus-PROVEN quantum; forward derivation OPEN.
IMPORTED	Result proved externally and used here; full citation given.
TESTABLE	Pre-registered prediction with explicit falsification protocol.
FLAGGED / TENSION	Verification check that did not pass cleanly; honestly registered.
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework’s scope.
OPEN	Recognized gap honestly registered; resolution path given where available.

§0. Abstract

We present the $T_Y \rightarrow X$ sector-conversion operator and its single, geometry-fixed nuclear prediction. The governing constant $A = \delta X \cdot \delta Y = (5/19)(7/23) = 35/437$ is derived from polyhedral curvature asymmetry (ZS-F2) and was fixed by the H_0 tension ($H_{0\text{local}}/H_{0\text{CMB}} = eA$) and the baryon-to-photon ratio η_B BEFORE any nuclear input — it is not tuned to lithium (§1.1).

Standard-physics statement: the ${}^3\text{He}(\alpha,\gamma){}^7\text{Be}$ radiative-capture rate receives a geometric correction $\delta\ln\eta_b = -\kappa^2 = -A/Q \approx -0.73\%$ on the ${}^7\text{Be}$ asymptotic normalization coefficient (ANC), equivalent to a fixed shift of the ${}^3\text{He}\text{--}{}^4\text{He}$ cluster-EFT short-range Wilson coefficient C_0 (§2.2).

Z-Spin-only prediction (TESTABLE): $S_{34}(\text{E})$ in the BBN Gamow window (0.1–0.5 MeV) is systematically lower by $\approx 0.7\text{--}1.5\%$, at the precision frontier of LUNA-MV (§4.2). This is an *interesting falsifiable hypothesis*: the predicted shift lies at or below current $\approx 4\%$ precision but within reach of LUNA-MV-class systematics ($\approx 1\text{--}2\%$), making the prediction testable in principle but not yet confirmed.

Immediate consistency test: a universal binding shift $-\text{dim}(\text{Z})\kappa^2$ would unbind ${}^8\text{B}$ ($\delta Q/Q = -401\%$) and destabilize ${}^9\text{C}$ (-44%); since both are observed (AME2020 [11]), the action must be bottleneck-selective (NC-T10.5). The master-equation slow eigenvalue is $\lambda_{\text{slow}} = -2A/Q = -\text{dim}(\text{Z})\kappa^2$ (§2.1). The structural anti-numerology Bayes factor is $\ln\Lambda = 7.2$ (very strong, not decisive; §6.5). The PRyMordial gate remains the principal caution: the physical ${}^7\text{Li}$ closure is coordinate-only (linear $+0.3\sigma$ vs nonlinear -3.3σ , Gate Li-2 FLAGGED). $C_0 \leftrightarrow \kappa^2$ **matching remains OPEN** (ZS-T11 programme, §9). Zero free parameters.

§1. Introduction

The cosmological lithium problem — the factor-3 excess of standard-BBN ${}^7\text{Li}/\text{H}$ over the Spite-plateau value — has resisted nuclear, astrophysical, and beyond-Standard-Model solutions for three decades. This paper does not claim to solve it. Instead it isolates a single, geometry-fixed correction to one reaction, states it in standard nuclear-physics language, and gives the one measurable number by which it can be falsified. ZS-T9 left two items OPEN: a forward operator carrying the conversion eigenvalue onto the ${}^7\text{Be}$ matrix element, and the O-F19.6 absolute scale. ZS-T10 establishes the operator and executes a PRyMordial stress-test that we report honestly, tension included.

§1.1. Prior Derivation of $A = 35/437$ and Its BBN Independence

A is not a lithium fit. It is the product structure $A = \delta\Box \cdot \delta^\vee = (5/19)(7/23) = 35/437$, computed from the face–vertex curvature asymmetry of the truncated-octahedron $\times$ truncated-icosahedron register geometry (ZS-F2, LOCKED), with no nuclear input. Its chronological priority is explicit:

Stage	Constraint that fixed A	Source / External reference
1. Geometry	$A = \delta\Box \cdot \delta^\vee$ from polyhedral curvature asymmetry	ZS-F2 (LOCKED)
2. Cosmology	$H_0^b/H_0^{\text{eMB}} = e^A$ (Hubble tension, 7.4%); tested against Riess et al. 2022 [12] (73.04 ± 1.04 km/s/Mpc) and Planck 2018 [13] (67.36 ± 0.54 km/s/Mpc): $e^A = 1.0834$, pull 0.06σ	ZS-F3, ZS-U4; [12,13] (external)
3. Baryogenesis	η_b via measure-projection weight; Planck 2018 [13] $+0.07\sigma$	ZS-F2 §9–10; [13] (external)

4. THEN BBN	Applied to ${}^7\text{Be}$ here, without adjustment	This work
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An external reader can verify that A was fixed by CMB/ H_0 and η_B data upstream of the present nuclear application: the lithium use is a prediction, not post-hoc tuning. The two external references [12,13] allow independent confirmation that e^A matches the observed H_0 ratio.

§1.2. Standard-Physics Translation (no Z-Spin vocabulary)

For readers outside the Z-Spin programme, the claim of this paper is: the ${}^3\text{He}-{}^4\text{He} \rightarrow {}^7\text{Be}$ radiative-capture amplitude receives a geometric correction $\delta \ln \gamma_\square = -A/Q \approx -0.73\%$ on the asymptotic normalization coefficient, arising from a Planck-scale sector-mediation constraint that is geometrically fixed (zero free parameters). In conventional terms this is a fractional shift of the ${}^7\text{Be}$ ground-state ANC, equivalently a shift of the leading short-range (contact) Wilson coefficient C_0 in halo/cluster EFT. It predicts a small, definite reduction of $S_{34}(E)$ at BBN energies and, under the linear sensitivity of Dent et al. [1], a downward ${}^7\text{Li}$ shift toward the Spite plateau. Whether that shift is the full lithium resolution is left open; the measurable content is the S_{34} reduction itself.

§2. The $T_{\gamma \rightarrow \epsilon}$ Operator

The ZS-Q7 inter-sector generator ($W_{AB} = \dim(B) \cdot A/Q$, the Fermi-golden-rule rate into sector B) over the coarse-grained sectors (X,Z,Y) factorizes as

$$\lambda(Q\lambda + \dim(Z)A)(Q\lambda + (\dim X + \dim Y + \dim Z)A)/Q^2 = 0,$$

giving $\lambda_{\text{slow}} = -\dim(Z)A/Q$ (the negative Z-bottleneck entry-rate, structural for arbitrary dimensions) and $\lambda_{\text{fast}} = -A$ (because $\dim X + Y + Z = Q$). Hence

$$T_{\gamma \rightarrow \epsilon} = -(\dim(Z) \cdot A/Q) \cdot P_{\square}^{\text{low}}, \quad \text{action level } \lambda_{\square}^{\text{low}} = -2A/Q = -70/4807,$$

lifted to DERIVED-CONDITIONAL by the Gorban–Radulescu–Zinovyev limiting-step theorem [3] (a multiscale linear network’s slow relaxation reduces to a single limiting step).

§2.1. The $\lambda_{\square}^{\text{low}} = -\dim(Z) \cdot \kappa^2$ Identity (DERIVED-CONDITIONAL)

With $\kappa^2 = A/Q$ (Register-Total Normalization, ZS-M6 §2.2 PROVEN): $\lambda_{\text{slow}} = -2A/Q = -2\kappa^2 = -\dim(Z) \cdot \kappa^2$ (exact rational $-70/4807$). Numerically identical to v1.0 — zero downstream change — but expressing the conversion eigenvalue through the PROVEN cross-sector coupling κ .

§2.2. What $T_{\gamma \rightarrow \epsilon}$ Is in Conventional Nuclear Language

In ${}^3\text{He}\text{--}{}^4\text{He}$ cluster EFT the ${}^7\text{Be}$ ground state is a shallow two-body bound state with binding momentum $\gamma_b = \sqrt{2\mu B}$ and ANC C . The operator acts as $\gamma_b \rightarrow \gamma_b(1 - \kappa^2)$, i.e., a fractional reduction of the binding momentum by $\kappa^2 = 0.728\%$. Because the external direct-capture cross-section scales with the ANC, this is equivalent, in standard EFT bookkeeping, to a shift of the leading short-range contact-term Wilson coefficient C_0 that reproduces the same $\delta C/C$. No new operator structure is introduced; the Z -Spin content is the fixed numerical value $\kappa^2 = A/Q$ of that shift. The explicit $C_0 \leftrightarrow \kappa^2$ matching requires a controlled renormalization-group running from Planck scale to nuclear scale and is the task of ZS-T11 (§9).

§2.3. Why the Action Is Bottleneck-Selective (theoretical basis, HYPOTHESIS-strong)

The ${}^8\text{B}/{}^9\text{C}$ test (§4.1) shows empirically that the operator cannot act as a universal binding rescaling. The theoretical reason follows from the same Gorban–Radulescu–Zinovyev limiting-step theorem [3] that lifts §2: in a multiscale linear network with well-separated rates, the reduction collapses the dynamics onto a SINGLE limiting (slowest) step. The action level of $T_{\gamma \rightarrow \epsilon}$ is precisely $\lambda \square^{\text{low}}$, i.e., that limiting step.

In the BBN mass-7 subnetwork the limiting step is ${}^7\text{Be}$ production and destruction (${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$; ${}^7\text{Be}(n, p){}^7\text{Li}$); ${}^8\text{B}$ enters only through the side branch ${}^7\text{Be}(p, \gamma){}^8\text{B}$, which is quantitatively negligible: at BBN temperatures ($T \sim 0.1\text{--}1.0$ MeV) the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction rate is $\sigma v \sim 10^{-27} \text{ cm}^3 \text{ s}^{-1}$, roughly $10^5\text{--}10^6$ times smaller than the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ rate at the same epoch [8]. The operator therefore acts on the ${}^7\text{Be}$ channel and not on ${}^8\text{B}$'s binding. This identification of “the limiting step” with ${}^7\text{Be}$ specifically is HYPOTHESIS-strong; its rigorous derivation from the BBN network Jacobian is the remit of ZS-T11 (§9).

§3. The ${}^7\text{Be}$ Cluster-EFT Scaffold and the $\delta \ln \gamma \square = -\kappa^2$ Foothold

${}^3\text{He}\text{--}{}^4\text{He}$: $\mu = 1601.6$ MeV, $B = Q_{\text{sep}} = 1.586$ MeV (AME2020 [11]), $\gamma_b = \sqrt{2\mu B}/\hbar c = 0.361 \text{ fm}^{-1}$. Since $\gamma_b \propto \sqrt{B}$, the operator gives $\delta \ln \gamma_b = \frac{1}{2} \delta \ln B = \frac{1}{2} \lambda_{\text{slow}} = -A/Q = -\kappa^2$ [HYPOTHESIS-strong]. The effective range parameter $\gamma_{\text{br}0} \approx 0.40\text{--}0.50$ is consistent with the ${}^3\text{He}\text{--}{}^4\text{He}$ scattering data compiled in Zhang, Nollett, Phillips (2020) [4]. The ANC modulation rate of ${}^7\text{Be}$ equals the PROVEN cross-sector coupling κ^2 ; the candidate Level-3 chain is ZS-S10 action $\rightarrow \kappa$ gauge bridge $\rightarrow \kappa^2$ ANC modulation $\rightarrow \delta \ln B_{7\text{Be}} = -2\kappa^2$ (E1 phase space). The E1 capture carries $E_\gamma^3 \propto Q^3$; with the PRyMordial forward slope $+0.969$ and amplification $B_{\text{tot}}/Q = 23.7$ the Dent sensitivity reconstructs to $+69$ (cf. Dent +81 [1]).

§4. PRyMordial Stress-Test

Standard-BBN baseline (PRyMordial small network, NACRE-II): ${}^7\text{Li}/\text{H} = 5.49 \times 10^{-10}$. $B_{7\text{Be}}$ enters via the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ detailed-balance Q -value and forward S -factor ($\partial \ln {}^7\text{Li} / \partial \ln \text{rate} = +0.969$).

Method	Response	⁷ Li outcome
LINEAR Dent +81	$\Delta \ln^7\text{Li} = -1.18$	$1.69 \times 10^{-10} (+0.3\sigma)$
NONLINEAR (Q ³)	eff. sens. +150	$0.62 \times 10^{-10} (-3.3\sigma)$
D/H, Y _□ (all runs)	$\Delta < 0.07\%$	decoupled ✓

The linear/nonlinear disagreement is registered as Gate Li-2 (FLAGGED): the physical ⁷Li closure is coordinate-only (NC-T10.1).

§4.1. Per-Nucleus Falsification Table and the ⁸B/⁹C Consistency Test (P-T10.1, TESTABLE)

If $\delta \ln B_{\square\square} = -2A/Q$, the implied Q-value shift is $\delta Q/Q = -2A/Q \cdot (B_{\square\square}/Q)$, with nucleus-specific amplification:

Nucleus	Cluster channel	$B_{\square\square}/Q$	$\delta Q/Q @ -2A/Q$
⁷ Be	³ He+ ⁴ He	23.7	−34.5%
⁶ Li	d+ ⁴ He	21.7	−31.6%
⁹ Be	⁸ Be+n	34.9	−50.9%
⁹ C	⁸ B+p	30.1	−44.0%
⁸ B	⁷ Be+p	275	−401% (unbinds)

Immediate consistency test (no new experiment needed). ⁸B is bound by only $S_p = 0.137$ MeV (AME2020 [11]), so a universal $-2A/Q$ binding shift implies $\delta Q/Q = -401\%$, which would completely unbind ⁸B; the same shift destabilizes ⁹C by -44% . Both nuclei are observed to exist. Therefore the operator CANNOT act as a universal binding rescaling — a literal universal action is immediately falsified by existing nuclear data — and must act only on the rate-limiting (bottleneck) channel, exactly the limiting-step selection of §2 (NC-T10.5). The ⁸B datum thus serves simultaneously as a falsification of the naïve reading and as positive evidence for the bottleneck-selective structure.

§4.2. The Z-Spin-Only Prediction and Its Experimental Programme (TESTABLE)

The bottleneck-selective reading applied to ⁷Be gives the one clean, standard-physics prediction of this paper: $\delta \ln \gamma_b = -\kappa^2 = -0.728\%$ on the ⁷Be ANC. For the external (peripheral) direct-capture reaction the low-energy S-factor is proportional to the ANC squared, $S_{34}(E) \propto C^2$ (ANC method [6,7]; halo-EFT leading order [5]), so $\delta \ln S_{34} = \delta \ln C^2$. In halo EFT the bound-state normalization gives $C^2 \propto \gamma_b/(1 -$

γ_{bro}), hence $d \ln C^2/d \ln \gamma_b = 1/(1 - \gamma_{bro})$, which runs from 1 at leading (zero-range) order to ≈ 2 with the ${}^3\text{He}\text{--}{}^4\text{He}$ effective range ($\gamma_{bro} \approx 0.40\text{--}0.50$, consistent with Zhang et al. 2020 [4]). Thus

$$\delta \ln S_{34}(E) \approx -0.7\% \text{ (LO) to } -1.5\% \text{ (with effective range), } \quad E \text{ in the BBN Gamow window (0.1--0.5 MeV).}$$

This is an *interesting falsifiable hypothesis*: it lies at or below current $\approx 4\%$ precision (LUNA activation at 127–169 keV [7]) but within reach of the LUNA-MV 3.5 MV underground accelerator now re-measuring ${}^3\text{He}(\alpha,\gamma){}^7\text{Be}$ over a wide energy range. Honest magnitude assessment: confirmation requires LUNA-MV-class systematics at $\approx 1\text{--}2\%$; we do NOT claim the shift is already measured (NC-T10.7).

Prediction	Test	Facility	Window
S_{34} low by $\sim 1\%$ ($\delta \ln \gamma = -\kappa^2$)	${}^3\text{He}(\alpha,\gamma){}^7\text{Be}$ S-factor, 0.1–0.5 MeV	LUNA-MV (Gran Sasso)	running ≥ 2019
${}^6\text{Li}$ ANC / S-factor shift	${}^2\text{H}(\alpha,\gamma){}^6\text{Li}$ direct capture	LUNA (Gran Sasso)	measured/ongoing
${}^8\text{B}$ unbinds under universal action	AME2020 ${}^8\text{B}/{}^8\text{C}$ exist $\rightarrow$ selective action	AME2020 [11]	immediate $\checkmark$

§5. The O-F19.6 Absolute-Scale Closure

Connection to the main result. O-F19.6 closes the second ZS-T9 OPEN item and is included because it fixes the absolute information scale of the SAME Z-bottleneck through which $T_\gamma \rightarrow \epsilon$ operates: both descend from the register equilibrium $p_{eq} = (3,2,6)/11$. The $\frac{1}{2}\ln 2$ entropy per $X \rightarrow Y$ conversion is the information-theoretic counterpart of the operator’s fixed $-2A/Q$ action level — it is what licenses treating that level as absolute rather than merely relative: the same geometric equilibrium that determines λ_{slow} also determines the per-transition entropy quantum $\frac{1}{2}\ln 2$, providing mutual consistency of the two results.

With $p_{eq} = (3,2,6)/11$ (L12 PROVEN) the modular Hamiltonian $K = -\ln p_{eq}$ has $X \rightarrow Y$ gap $\Delta K = -\ln 2$; the half-gap $\frac{1}{2}\ln 2 = \psi_{KMS}$ reproduces $\tanh(2\psi_{KMS}) = 3/5$ (Theorem F19.6). The Type II $_{\infty}$ trace weight being fixed by p_{eq} , $\Delta S_{gen} = \frac{1}{2}\ln 2 = 0.5$ bit is ABSOLUTE, with register constant $\ln Q = \ln 11$. DERIVED-CONDITIONAL on the $(3,2,6) \leftrightarrow$ degenerate-clock identification [2].

§6. Anti-Numerology Monte Carlo

- Part A (500k):** $\lambda_{slow} = -\dim(Z)A/Q$ in 100.0000% of random configs (max err 1.8×10^{-15}). Structural identity, not fit.
- Part C (500k):** Arbitrary-rate chains reproduce $-2A/Q$ only 0.54% — the Fermi-golden-rule rate law is required.

- **Part B (2M + 414 quanta):** Value-fit selectivity 3.9% (modest); at fixed $-2A/Q$ only B7Be lands (others +6 to +20 σ).

§6.4. Structural vs Statistical Anti-Numerology

$-2A/Q$ is a fixed eigenvalue, not a fitted parameter, so the statistical fit-test (Part B, 3.9%) is the wrong instrument. The operative evidence is structural: Part A (100% structural identity) and Part C (0.54% rate-law necessity). A deterministic eigenvalue carries no probability factor; forming a “combined probability” by multiplying in a zero is a numerology-adjacent error and is explicitly NOT done here.

§6.5. Bayes Factor for the Structural Evidence (DERIVED-CONDITIONAL)

Explicit hypotheses. H_0 (null): the action level $-2A/Q$ landing on the ${}^7\text{Be}$ channel is a chance coincidence of an arbitrary inter-sector rate structure. H_1 (Z-Spin): the structural derivation predicts $-2A/Q$ (the master slow eigenvalue) acting on the mass-7 bottleneck. The Bayes factor is $\Lambda = P(\text{data}|H_1)/P(\text{data}|H_0)$. Under H_1 the prediction is deterministic, $P(\text{data}|H_1) \approx 1$. Under H_0 the data require two independent coincidences: an arbitrary rate structure reproducing $-2A/Q$ (MC Part C, $p^c = 0.0054$) AND that structure selecting the ${}^7\text{Be}$ channel among the seven decoupled binding channels ($p^{\text{chan}} = 1/7$). Assuming independence: $P(\text{data}|H_0) = p^c \cdot p^{\text{chan}} \approx 7.7 \times 10^{-4}$, giving $\Lambda \approx 1296$, $\ln \Lambda = 7.2$.

Honest caveats. This is an order-of-magnitude likelihood ratio, not a full Bayes factor: (i) the independence of value- and channel-coincidences is assumed; (ii) $P(\text{data}|H_1) = 1$ ignores the prior plausibility of H_1 itself; (iii) p^c and p^{chan} are Monte-Carlo point estimates. On the Jeffreys scale $\ln \Lambda = 7.2$ is “very strong” ($\ln \Lambda > 5$) but NOT “decisive” ($\ln \Lambda > 10$). Status: DERIVED-CONDITIONAL on those assumptions.

§7. Falsification Gates

Gate	Condition	Status
Li-1	PRyM reproduces Dent +81 at linear order	PASS (+69 LO)
Li-2	Nonlinear B7Be injection matches linear closure	FLAG / TENSION
Li-3	D/H and $Y_{\square}$ decoupled ($\Delta < 0.07\%$)	PASS
P-T10.1	${}^8\text{B}$ unbinds / ${}^9\text{C}$ destabilized under universal action — FALSIFIED by AME2020	PASS-consistency (selective)
S34-1	$S_{34}(0.1\text{--}0.5 \text{ MeV})$ low by $\sim 1\%$ ($\delta \ln \gamma_{\square} = -\kappa^2$); interesting falsifiable hypothesis at LUNA-MV precision	TESTABLE (LUNA-MV)

S-T10.1	Cluster-EFT $C_0 \leftrightarrow \kappa^2$ matching (explicit RG running from Planck to nuclear scale)	OPEN (ZS-T11)
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§8. Non-Claims

- **NC-T10.1.** The ${}^7\text{Li}$ binding-channel closure is coordinate-only; Gate Li-2 does not validate it as a physical mechanism.
- **NC-T10.4.** No new free parameter; $A = \delta \square \delta^\vee$, Q , $\dim Z$, $\kappa^2 = A/Q$ all LOCKED.
- **NC-T10.5.** The operator acts bottleneck-selectively, NOT as a universal binding rescaling (${}^8\text{B}/{}^9\text{C}$ test forbids the latter).
- **NC-T10.6.** The cross-corpus invariant is $\dim(Z) = 2$, NOT the rational $-2A/Q$.
- **NC-T10.7.** The predicted S_{34} shift ($\sim 1\%$) is not claimed to be already measured; it is a target for LUNA-MV-class precision.
- **NC-T10.8 (v2.0).** $C_0 \leftrightarrow \kappa^2$ matching is NOT claimed in this paper; the identification requires an explicit RG running calculation from the Planck-scale Z-Spin gauge bridge ($\kappa^2 = A/Q$, ZS-S10) to the nuclear EFT contact term (C_0). That calculation is the deliverable of ZS-T11.

§9. Conclusion, $\dim(Z)=2$ Convergence, and the ZS-T11 Programme

ZS-T10 isolates one geometry-fixed, standard-language prediction — $\delta \ln \gamma_b = -\kappa^2$ on the ${}^7\text{Be}$ ANC, hence S_{34} low by $\sim 1\%$ — with A fixed by CMB/ H_0 [12,13] long before any nuclear use, and an immediate ${}^8\text{B}/{}^9\text{C}$ consistency test. The cross-corpus signal is the cardinal-2 invariant $\dim(Z) = 2$, not the rational $-2A/Q$ (NC-T10.6):

Appearance	Quantity	Status
ZS-Q7 slow eigenvalue	$-2A/Q = -\dim(Z)\kappa^2$	PROVEN / DERIVED
ZS-U10 Schwinger coeff.	$C \square = \dim(Z)/(4\pi) = 2/(4\pi)$	DERIVED
ZS-F19 KMS rapidity	$\frac{1}{2}\ln 2$, $\dim(Z) = 2$ channels	DERIVED
ZS-A12 Nelson–Kosterlitz	universal-jump integer 2	DERIVED

ZS-T11 programme (path to DERIVED). The single calculation that would lift every HYPOTHESIS-strong tag in ZS-T10 to DERIVED is the explicit matching of the ${}^3\text{He}$ – ${}^4\text{He}$ cluster-EFT short-range contact term C_0 to $\kappa^2 = A/Q$. This requires three steps: (1) derive the matching condition $C_0(\kappa^2)$ via controlled RG running from the ZS-S10 Planck-scale gauge bridge ($t^2 = \kappa^2 M^{\text{p}2}$) to the nuclear

EFT scale; (2) compute the resulting ANC and $S_{34}(E)$; (3) compare to the Zhang–Nollett–Phillips (2020) [4] determination. Agreement would promote $\delta \ln \gamma_b = -\kappa^2$ from HYPOTHESIS-strong to DERIVED and make the S_{34} prediction a parameter-free number. That calculation is the deliverable of ZS-T11. Until it is performed, the $C_0 \leftrightarrow \kappa^2$ identification remains OPEN (Gate S-T10.1), and the S_{34} prediction remains an interesting falsifiable hypothesis at the LUNA-MV precision frontier.

Acknowledgements & Code Availability

PRyMordial (Burns, Tait, Valli [9]) was used under its public license for the BBN runs. All symbolic, Monte-Carlo, EFT, O-F19.6, $^8\text{B}/^9\text{C}$ -consistency, Bayes-factor, and PRyMordial-gate scripts (32 checks, fixed random seeds) reproduce every figure herein and are available on request. No new free parameters were introduced.

Appendix A. Numerical Inputs

Quantity	Value	Source
$A = \delta \square \cdot \delta \nu$	$(5/19)(7/23) = 35/437$	ZS-F2 (geometry)
$\kappa^2 = A/Q; \lambda \square^{\text{loW}}$	$35/4807; -70/4807$	ZS-M6; ZS-Q7
$\delta \ln \gamma \square = -\kappa^2$	$-0.00728 \text{ } (-0.73\%)$	This work §3
$\gamma \square_{\text{r0}} \text{ } (^3\text{He}\text{--}^4\text{He eff. range})$	$0.40\text{--}0.50$	Zhang et al. 2020 [4]
$\delta \ln S_{34} \text{ (predicted)}$	$-0.7\% \text{ to } -1.5\%$	This work §4.2
$^8\text{B } S \square; B \square_o \square/Q$	$0.137 \text{ MeV}; 275$	AME2020 [11]
$^9\text{C } S \square; B \square_o \square/Q$	$1.299 \text{ MeV}; 30.1$	AME2020 [11]
Bayes factor $\ln \Lambda$	$7.2 \text{ (very strong)}$	This work §6.5
PRyM baseline $^7\text{Li}/\text{H}$	5.49×10^{-10}	This work
H_0 ratio e^A (external check)	$1.0834 \text{ vs Riess } 73.04, \text{ Planck } 67.36$	[12,13]

Appendix B. Review-Audit Record (v1.0 → v2.0)

All corrections adopted under the no-deletion rule. Items from v1.1, v1.2, v1.3 preserved; v2.0 adds items 11–14.

#	Issue found	Version	v2.0 resolution
1	Universal $B_{\text{O},Q}$ ‘20–30’ FALSIFIED (${}^8\text{B}=275$)	v1.1	§4.1 per-nucleus table
2	$\delta \ln \gamma = -\kappa$ FACTUAL ERROR (correct: $-\kappa^2$)	v1.1	§3 $\delta \ln \gamma = -\kappa^2$
3	$2.3\% \times 0\% \times 0.54\% = 0$ LOGIC ERROR	v1.1	§6.4 structural vs statistical; no product with zero
4	$-2A/Q$ convergence across U10, F19 OVERCLAIM ($U10 = \dim(Z)/4\pi$)	v1.1	§9 $\dim(Z) = 2$ convergence
5	S_{34} prediction magnitude $\sim 1\%$ not yet measured	v1.2	NC-T10.7; stated honestly
6	${}^9\text{C}$ ‘disappears’ (${}^9\text{C}$ is -44% , not full unbinding)	v1.2	§4.1 ${}^8\text{B}=-401\%$ unbinds; ${}^9\text{C}=-44\%$
7	$\ln \Lambda = 7.2$ ‘decisive’ ($\neq$ decisive, which requires >10)	v1.2	§6.5 stated as 7.2 (>5 , not >10)
8	${}^6\text{Li}$ reaction: ${}^6\text{Li}(p,\gamma) \rightarrow$ correct channel ${}^2\text{H}(\alpha,\gamma){}^6\text{Li}$	v1.2	§4.2 corrected channel
9	S_{34} sensitivity ‘1–2’ uncited hand-wave	v1.3	§4.2 $S \propto \text{ANC}^2$ [6,7]; $1/(1-\gamma r_0)$ basis
10	Selectivity theory only empirical (${}^8\text{B}$), no theory	v1.3	§2.3 limiting-step $\rightarrow$ ${}^7\text{Be}$ bottleneck (with quantitative ${}^8\text{B}$ rate ratio)
11	$\ln \Lambda$ definition: H_0/H_1 undefined; product unjustified	v1.3	§6.5 explicit H_0/H_1 + independence caveat
12	§5 O-F19.6 disconnected from main result	v1.3	§5 shared $p_{e\gamma}=(3,2,6)/11$ link sentence
13	H_0 external citation absent for A’s BBN independence claim	v2.0	§1.1 table row 2: Riess 2022 [12], Planck 2018 [13] added; App. A last row
14	$\gamma r_0 \approx 0.4-0.5$ uncited	v2.0	§3 ‘consistent with Zhang et al. 2020 [4]’ added; App. A row added
15	$C_0 \leftrightarrow \kappa^2$ OPEN scope not clearly bounded	v2.0	§2.2 + NC-T10.8 + §9 ZS-T11 programme: RG running Planck $\rightarrow$ nuclear EFT explicitly stated as OPEN
16	Reference [7] duplicate (Costantini & Bemmerer both [7])	v1.3 $\rightarrow$ v2.0	References renumbered: Igamov [6], Costantini [7], Bemmerer [8], Adelberger [9], Burns [10], Fields [11], AME2020 [12]; Riess [13], Planck [14] added

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- [14] Planck Collaboration, A&A 641, A6 (2020). [$H_0^{\text{CMB}} = 67.36 \pm 0.54$ km/s/Mpc; external check for A’s BBN independence]

Version History

v1.0 (Mar 2026): Operator $T_\gamma \rightarrow \epsilon$; PRyMordial gate with registered nonlinear ${}^7\text{Li}$ tension (Li-2).

v1.1 (Mar 2026): $\lambda\Box^{\text{low}} = -\dim(Z)\kappa^2$; $\delta\ln\gamma\Box = -\kappa^2$; per-nucleus P-T10.1; structural-vs-statistical §6.4; $\dim(Z) = 2$ convergence. Four review proposals corrected (App. B #1–4).

v1.2 (Mar 2026): First external-reviewer revision. Adds §1.1 A’s chronological priority; §1.2 standard-physics translation; §2.2 ANC/Wilson coefficient restatement; §4.1 ${}^8\text{B}/{}^9\text{C}$ consistency test; §4.2 S_{34} prediction + LUNA-MV programme; §6.5 Bayes factor $\ln\Lambda = 7.2$; §9 ZS-T11 programme. Three prescriptions corrected (App. B #5–8).

v1.3 (Mar 2026): Second external-reviewer revision. Adds §2.3 limiting-step $\rightarrow {}^7\text{Be}$ theoretical basis; $\gamma\Box r_0 = 0.40\text{--}0.50$ ANC sensitivity in §4.2; explicit H_0/H_1 + independence caveats in §6.5; shared- $p_{e\gamma}$ sentence in §5. Four items corrected (App. B #9–12).

v2.0 (Mar 2026): Consolidation revision closing four legibility gaps. Adds: H_0 external references [13,14] and $\gamma\Box r_0$ citation [4] in §1.1 and §3; NC-T10.8 bounding the $C_0 \leftrightarrow \kappa^2$ OPEN scope; ZS-T11 RG-running requirement made explicit in §2.2 and §9; reference renumbering to fix v1.3 [7]-duplicate; App. B extended to 16 items. S_{34} prediction repositioned explicitly as ‘interesting falsifiable hypothesis’ at LUNA-MV precision frontier. No numerical result changed; zero new free parameters.